Ariel Gjaci

Postdoctoral Researcher | Multimodal Generative Models | Diffusion Transformers

ariel.gjaci@iit.it | +393483776041 | Genoa, Italy

Website | LinkedIn | GitHub | Google Scholar

Postdoctoral researcher with a PhD in Artificial Intelligence and Robotics. Develops multimodal generative models for human motion and spatial audio, combining diffusion transformers, representation learning, dataset development, and experimental evaluation in Python and PyTorch.

TECHNICAL SKILLS

Generative Models
Diffusion transformers; GANs; Multimodal conditioning

Learning and Evaluation
Representation learning; Domain generalization; Generative model evaluation

Programming
Python; C/C++; MATLAB/Simulink

Frameworks
PyTorch; Scikit-learn; TensorFlow

Tools and Training
Git; Linux; SLURM; Data parallelism

EDUCATION

Ph.D. Artificial Intelligence and Robotics
University of Genoa
Nov 2021 - May 2025

Research collaboration with King's College London.

M.Sc. Robotics Engineering
University of Genoa
2018 - 2021

B.Sc. Biomedical Engineering
University of Genoa
2014 - 2018

LANGUAGES

Italian (native)
Albanian (native)
English (fluent)

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher
Italian Institute of Technology (IIT)
Feb 2025 - Present

- Developing few-shot room impulse response generation from audio and depth, with counterfactual training for reverberation control (ongoing research).
- Built Eigenmike 64 processing pipelines for beamforming, binaural rendering, and acoustic mapping on 3D room point clouds.
- Implemented audio/video synchronization and 360-degree acoustic overlays for spatial-audio recordings.

PhD Researcher
University of Genoa
Nov 2021 - May 2025

- Designed ALaDiT, a diffusion transformer conditioned on speech, text, and cultural representations for 3D gesture generation (SICAGE, ECCV 2026).
- Built a video-to-3D-motion pipeline and released TED4C-L: 106.45 hours of aligned motion, audio, and text from 764 speakers.
- Evaluated motion quality, diversity, and speech alignment through baseline comparisons and controlled ablations; released training and evaluation code.

Software Engineer
AKKA Technologies
May 2021 - Oct 2021

- Developed Python backend software for aircraft panels and presented a working demo to stakeholders.

SELECTED PUBLICATIONS

SICAGE: Speaker-Independent Culture-Aware Gesture Generation using TED4C-L Dataset. European Conference on Computer Vision (ECCV), 2026.

Labeling Sentences with Symbolic and Deictic Gestures via Semantic Similarity. IEEE RO-MAN, 2024.

Towards Culture-Aware Co-Speech Gestures for Social Robots. International Journal of Social Robotics, 2022.

Full publication list

TEACHING AND SERVICE

Teaching assistant (C++, Android/Kotlin); ECCV workshop co-organizer; reviewer for AAAI, ECCV, ACM Multimedia, IROS, ICMI, and IJSR; IEEE RO-MAN 2024 session chair.